Contents

Thermal representation as a decision variable in heat-adaptive tourism opportunity screening: evidence from a Madrid pilot **2**
Abstract . . . 2

1. Introduction **3**

2. Methods **5**
2.1 Study design and study area . . . 5
2.2 Data sources and preprocessing . . . 6
2.3 Simple-proxy baseline . . . 7
2.4 Physically based thermal-exposure modelling . . . 7
2.5 Thermal-method comparison . . . 8
2.6 Uncertainty and robustness treatment . . . 9
2.7 Constraint-first tourism opportunity screening . . . 9
2.8 Conventional baseline and decision scenarios . . . 10
2.9 Validation strategy and claim boundaries . . . 10

3. Results **11**
3.1 Thermal-method sensitivity . . . 11
3.2 Effects on tourism-opportunity screening . . . 12
3.3 Robustness and traceability . . . 13

4. Discussion **14**
4.1 Thermal representation is decision-relevant . . . 14
4.2 Screening before routing: a distinct decision-support layer . . . 15
4.3 Explicit non-recommendation and traceability . . . 15
4.4 Robustness, uncertainty, and appropriate confidence . . . 16
4.5 Contribution and transferability . . . 17

5. Limitations and scope conditions **17**
5.1 Physical-model validation boundary . . . 17
5.2 Spatial and temporal scope . . . 18
5.3 Input and uncertainty limitations . . . 18
5.4 Tourism and accessibility limitations . . . 19
5.5 Operational-data and retrospective-alignment limitations . . . 19
5.6 Scope conditions and transferability . . . 20
5.7 Future work . . . 20

6. Conclusion **20**

Declarations **21**

References **22**

Figure captions **24**

Table captions **25**

Thermal representation as a decision variable in heat-adaptive tourism opportunity screening: evidence from a Madrid pilot

Prepared for submission to **Tourism Management Perspectives** (primary target).

Authors: [AUTHOR NAME(S) TO VERIFY]

Affiliations: [AFFILIATION(S) TO VERIFY]

Corresponding author: [NAME TO VERIFY], [EMAIL TO VERIFY]

ORCID: [ORCID(S) TO VERIFY]

Abstract

Extreme heat is a time- and place-sensitive management problem for urban tourism, yet broad climate-suitability assessment operates at a coarse scale, while downstream thermal routing and heat-adjusted accessibility presuppose the candidate set, leaving open the upstream question of which tourism opportunities should remain feasible candidates at a given hour. This study develops a constraint-first, uncertainty-aware architecture for screening urban tourism opportunities under heat and tests how much the choice of thermal representation matters. On a documented extreme-heat day in Madrid, across 27 tourism assets at three times of day, feasibility was assessed two ways: with a simple operational proxy combining ambient air-temperature hazard thresholds and nearby tree-presence information, and with physically based SOLWEIG/UTCI modelling. Changing the thermal representation reclassified one third of outdoor asset-time observations (33.3%, 14 of 42), in both directions. Relative to a conventional nearest-open baseline, constraint-first screening changed the feasible-alternative set in 7 of 8 decision scenarios, the nearest-open option failed screening in 3 of 8, and one scenario returned an explicit no-defensible-alternative outcome when no candidate qualified. The analysis evaluates decision sensitivity to thermal-method choice rather than the accuracy of either method against ground truth. The findings indicate that thermal representation is a substantive modelling choice in heat-adaptive tourism decision support, and that a constraint-first architecture can expose that sensitivity while keeping uncertainty, evidence, and exclusion logic explicit and complementing downstream routing and accessibility methods. Results are limited to a single Madrid pilot without direct field validation of Tmrt/UTCI or observed tourist behaviour.

Keywords: Urban tourism; Extreme heat; UTCI; Tourism decision support; SOLWEIG; Uncertainty-aware decision-making

---

1. Introduction

In the tourism cities of Southern Europe, extreme-heat episodes increasingly overlap with the summer season and with the daytime, outdoor conditions under which much urban tourism takes place. Open plazas, monument forecourts, gardens, and the walking segments between attractions become heat-exposed during these hours, and aggregate studies show that urban activity contracts and shifts as temperatures rise (Renninger & Cabrera, 2026). For destination management this is an operational problem rather than only a long-run climatic one: on a given afternoon, some outdoor opportunities become uncomfortable or unsuitable while others, and most indoor options, remain usable, and visitors and the organisations that guide them must decide among these options in near-real time. Yet heat governance has developed largely around residents and public-health warning systems, and frameworks for embedding tourism in heat action planning are only now being articulated (Scott, 2026; OECD, 2026). Southern-European destinations are already adapting in practice (Ketter & Farkash, 2026), and destination demand and development are increasingly shaped by these conditions (Gössling & Scott, 2025). Critically, the exposure of tourism to heat is spatially uneven at the intra-urban scale (Camatti et al., 2024), which is exactly the scale at which visitors experience comfort in a historic core (Karimi & Mohammad, 2022). The practical decision is opportunity-level and time-specific: which of the attractions in front of a visitor this afternoon should remain on the table.

The dominant way to characterise tourism and climate has been the composite climate index. Instruments such as the Tourism Climate Index and the Holiday Climate Index combine several weather variables into an aggregate suitability score, and they are well suited to comparing destinations or seasons at a broad scale (Scott et al., 2016; Kovács et al., 2025). These indices remain useful for their intended purpose, but two features limit their reach for opportunity-level screening. First, their outputs are known to depend on the choice of component weights and thresholds, so the same conditions can yield different suitability readings under different, defensible parameterisations (Dubois et al., 2016) — a property shared with weighted multi-criteria site-suitability approaches more broadly (Bunruamkaew & Murayama, 2011). Second, and more fundamentally here, they are designed for destination- or season-scale characterisation, not for resolving which individual microsite is feasible at which hour. A city-scale or daily suitability summary does not distinguish a sun-exposed plaza from a shaded garden a few hundred metres away, nor a noon exposure from a late-afternoon one. Screening an individual opportunity requires a per-site, per-hour eligibility judgement — is this place usable now, and if not, what nearby option is — which an aggregate score is not constructed to provide. The limitation is one of spatial and temporal scale and of decision purpose, not of validity.

Physically based thermal modelling resolves part of what an aggregate or air-temperature-based summary cannot. Mean radiant temperature (Tmrt) represents the combined short- and long-wave radiant load on a person, and radiation-resolving models of the SOLWEIG class compute it at pedestrian scale from building and vegetation geometry and the solar path (Lindberg et al., 2008); combined with air temperature, humidity, and wind, it yields the Universal Thermal Climate Index (UTCI), a modelled index of heat stress on a standardised body (Bröde et al., 2012). Such models capture the shade- and geometry-driven heterogeneity that ambient temperature and simple vegetation counts do not, and they have been used for city-scale comfort mapping and design (Ding et al., 2024), with a growing literature cautioning that surface- or air-temperature summaries are not interchangeable with human-centred heat stress (Wang et al., 2026). To date, however, these radiation-resolving models have mostly served design, planning, and mapping; they are rarely wired transparently into an auditable, opportunity-level tourism decision, so their information reaches the screening stage only indirectly, if at all. It is important to be exact about what this buys. These approaches provide a different representation of the thermal environment, resolving additional physical dimensions relevant to pedestrian exposure; they are not, on their own and without in-situ measurement, a validated ground truth against which a simpler method can be called wrong (Gál & Kántor, 2019). The question this raises for tourism decision-support is therefore not which thermal method is correct, but whether the choice of thermal representation materially changes a tourism-screening decision — a question that can be answered by comparison even where neither representation has been field-validated.

A fast-moving strand of heat-mobility research has begun to connect thermal exposure to movement, and it defines the boundary of the present study most sharply. Thermal routing takes an origin and a destination as given and finds the path of least thermal exposure between them (Buo et al., 2026; Wolf et al., 2025). Thermal-adjusted accessibility takes a destination set as given and measures how many destinations remain reachable under heat-adjusted travel (Aydin et al., 2026), and related work maps sidewalk-level heat risk along pedestrian networks (Colaninno et al., 2025) or examines whether cool refuges are equitably reachable (Mombelli et al., 2025). These methods are advancing quickly and are complementary to what follows; they operate, however, at a downstream decision stage. Each presupposes the candidate set — the destinations or refuges whose paths or reachability are then evaluated. The upstream management question is distinct: which tourism opportunities should remain eligible candidates at a given hour under thermal, operational, accessibility, and evidence constraints, before any route is planned or any reachability computed. That screening step, and the surviving candidate set it produces, is what a downstream routing or accessibility method would consume.

Among the closest approaches identified in a targeted review of this literature, we found that routing, accessibility, thermal mapping, and composite climate suitability are each addressed separately, and none screens individual tourism opportunities by explicit physical-thermal, accessibility, and evidence constraints at a specific hour while keeping hazard, exposure, confidence, and evidence traceably distinct. The present study addresses that screening stage. It develops a constraint-first, uncertainty-aware architecture for screening urban tourism opportunities under heat — one that evaluates each opportunity against an ordered sequence of hard constraints, records a single machine-readable reason for every exclusion, keeps the thermal state separate from the confidence in the decision, and returns a set of surviving alternatives or, where nothing qualifies, an explicit no-recommendation outcome rather than a forced choice. In treating uncertainty as a first-class, categorical property of each decision rather than folding it into a single composite score, the architecture draws on the tradition of uncertainty-aware spatial decision-support (Huck et al., 2025; Sargazi et al., 2025). Because the architecture places the thermal input in an explicit and swappable position, it also supports a direct empirical test of how much that input matters: the study compares a simple operational proxy — combining ambient air-temperature hazard thresholds with nearby tree-presence information — against a physically based (SOLWEIG/UTCI) configuration, and against a conventional nearest-open recommender, on a single documented extreme-heat day in central Madrid. The contribution is thus an applied, reproducible decision-support architecture together with a case-based demonstration of whether thermal-method choice is decision-relevant; it does not rest on either thermal method being validated, and it makes no claim about tourist behaviour, flows, or outcomes. The study is guided by three research questions:

- **RQ1 (method sensitivity).** Does physically based (SOLWEIG/UTCI) thermal modelling materially alter tourism-feasibility classifications relative to the simple air-temperature-and-tree-presence proxy, and is any difference physically interpretable?
- **RQ2 (decision-support value).** Does constraint-first, heat-aware screening change the set of feasible alternatives relative to a conventional nearest-open baseline, and does it decline to recommend when nothing accessible qualifies?
- **RQ3 (robustness and traceability).** Do those decisions remain auditable and stable under the tested uncertainty while preserving explicit no-recommendation outcomes?

---

2. Methods

2.1 Study design and study area

This study is a single-city, single-day pilot designed to test whether the choice of thermal-exposure method changes heat-adaptive tourism screening decisions, and to demonstrate a transparent decision architecture in which any such difference is made explicit. It deliberately trades breadth for auditability, using a compact study area, a fixed set of real tourism assets, and a few fixed hours.

The study area is a rectangular box of approximately 3.5 km² in central Madrid, spanning the Paseo del Prado museum corridor, the Puerta de Alcalá and Cibeles plazas, the Atocha transit hub, and a western slice of the Retiro park interior (the Prado–Retiro–Atocha box). The rectangle is pinned to named, independently verifiable landmarks so that it can be reproduced without any proprietary boundary file, and was drawn to include a cross-section of adaptation conditions — open monumental plazas, an indoor museum corridor, a transit interchange, and shaded gardens — while remaining small enough to keep every asset individually inspectable (Fig. 1a).

Within this area we curated 27 real, named tourism assets from OpenStreetMap (13 indoor, 14 outdoor), each resolved to a specific OSM feature identifier. The set was hand-selected to span the adaptation conditions above rather than sampled at random; it is therefore a purposive pilot, not a representative sample, and no inference to the full population of Madrid tourism sites is drawn from it.

The study day is 21 August 2023, which falls inside an extreme-heat episode (20–25 August 2023) formally designated by the Spanish State Meteorological Agency

(AEMET) for the Madrid region. Using an officially designated episode, rather than a self-selected hot day, anchors the “extreme heat” framing to an external authority. The day is treated as a representative extreme-heat case only; it is not claimed to represent Madrid climatologically, and no seasonal or multi-day generalisation is made from it. Three fixed local hours are analysed — 12:00, 15:00, and 18:00 (CEST) — with observed air temperatures of 34.2, 38.8, and 40.5 °C respectively. These hours were fixed by the study design before any data were retrieved.

The primary unit of analysis is the tourism asset × timestamp pair. Thermal-method comparison is performed on the 42 outdoor pairs (14 outdoor assets × 3 hours); indoor assets bypass the outdoor thermal model by construction (Section 2.4). The individual tourist is explicitly not a unit of analysis, and no quantity in this study describes tourist behaviour, choice, or flow.

2.2 Data sources and preprocessing

All inputs are open data with documented provenance (Table 1). We distinguish three categories: observed meteorological data, modelled or derived physical inputs, and contextual tourism data.

Observed meteorological data. Hourly air temperature, relative humidity, wind speed, and pressure for 21 August 2023 were taken from the Madrid/Barajas station (WMO 08221, ICAO LEMD), the nearest station with a genuine (non-modelled) hourly observation record. The station lies approximately 9 km north-east of the study area on an airport apron; its readings are used as the best available real, sub-daily anchor for the regional heat signal, not as an estimate of on-site air temperature at each asset, and each record carries a note quantifying the station’s agreement with AEMET’s in-park Retiro reference (about +0.5 °C at daily maximum). A co-located Retiro hourly series was sought but found to be reanalysis-interpolated rather than observed, and was not used. The meteorological warning thresholds and the episode designation are drawn from official AEMET publications.

Modelled and derived physical inputs. The physical thermal model (Section 2.4) is driven by three-dimensional urban geometry from Spanish national LiDAR products (IGN/CNIG): a 5 m digital elevation model and 2.5 m building- and vegetation-height layers, each clipped to the study area. Global horizontal irradiance, which the Barajas hourly archive does not record, was estimated with a clear-sky radiation model (the Ineichen–Perez formulation) evaluated at the study-area centroid and true timestamps, a substitution justified by the archive’s own clear-sky weather code at all three hours. This irradiance is an estimated, not observed, input and is treated as such throughout.

Contextual tourism data. Asset locations, categories, tree points, and park/garden polygons are from OpenStreetMap (ODbL). Opening hours were harvested from OpenStreetMap tags where present (11 of 27 assets) and otherwise filled from documented institutional schedules (16 of 27), each recorded with its source and an evidence-completeness flag. One temporal-alignment limitation is material and is stated explicitly: the OpenStreetMap snapshot and the institutional opening hours were retrieved in 2026 and applied retrospectively to the 2023 study date. This assumes the pilot area's urban form and operating schedules did not change materially over that window — reasonable for a stable, protected city core, but not verified — and it is carried forward as a permanent limitation.

2.3 Simple-proxy baseline

The comparison baseline is a constraint-first, open-data feasibility classification that uses no radiative, surface-temperature, or shadow information. It combines exactly two inputs through a decision rule. It is not a land-surface- temperature product, a satellite surface-temperature proxy, a canopy or shade model, or a shadow simulation; no such quantity enters it.

The first input is an ambient air-temperature hazard band. The observed hourly air temperature (Section 2.2) is classified against AEMET's official Meteoalerta maximum-temperature civil-protection thresholds for the Madrid metropolitan zone (warning levels at 36, 39, and 42 °C), yielding bands LOW, ELEVATED, SEVERE, and EXTREME. Under this scale the three study hours fall in LOW (12:00), ELEVATED (15:00), and SEVERE (18:00). This is a meteorological hazard classification of a station reading against a warning scale; it is not, and is never read as, a thermal-comfort index.

The second input is an exposure band derived from the local count of OpenStreetMap `natural=tree` points within the asset's real extent — a 50 m radius buffer for point-type assets, or the asset's own park/garden polygon buffered by 15 m for the eight area-type assets. The tree count is classified into LOW, MODERATE, or HIGH exposure by tercile of the pilot's own 14-outdoor-asset tree-count distribution (tercile cut-points of 0.33 and 3.67 trees), where HIGH denotes the fewest trees (poorest shade availability). These terciles are an explicit within-sample relative grade, not a transferable threshold, because no published tree-count shade- sufficiency standard exists to cite. The tree count is a presence proxy for local shade availability; it does not measure canopy cover, shadow geometry, or sun position.

The two bands are combined by a first-matching-rule decision tree, with no weighted score. For outdoor assets: an EXTREME hazard band yields NOT RECOMMENDED; SEVERE yields NOT RECOMMENDED where exposure is HIGH and otherwise FEASIBLE WITH CONDITIONS; ELEVATED yields FEASIBLE WITH CONDITIONS; and LOW yields FEASIBLE, or FEASIBLE WITH CONDITIONS where exposure is HIGH. Indoor assets bypass the exposure input. This baseline approximates pedestrian heat load with two inexpensive surrogates — an ambient hazard signal and local tree presence — and represents the simplest defensible open-data screening architecture available before any physical modelling; it does not measure human thermal comfort.

2.4 Physically based thermal-exposure modelling

The physical comparator replaces the two-surrogate baseline with a radiation- resolving estimate of pedestrian-level heat load. Mean radiant temperature (Tmrt) was modelled with SOLWEIG (Lindberg et al., 2008), a three-dimensional radiant-flux model, and combined with meteorological forcing to derive the Universal Thermal Climate Index (UTCI). We used the official standalone `solweig` package (version 0.1.0b92), with

UTCI computed by the package's own built-in module from the modelled Tmrt and the same observed air temperature, humidity, and wind. Computation is at a 2.5 m working resolution — the native resolution of the building- and vegetation-height layers — with the digital elevation model resampled onto that grid. Each hour was run as an independent single-timestep calculation. Model parameters were left at the package's documented defaults (anisotropic sky, deciduous leaf-on transmissivity appropriate to the August date, standing-posture human geometry, default material set), with no parameter tuned to a target result and no custom land-cover grid supplied.

Tmrt represents the combined short- and long-wave radiant environment experienced by a standardised human body; UTCI combines that radiant environment with air temperature, humidity, and wind into a model-derived thermal-stress index calibrated by Bröde et al. (2012). UTCI is a model output, not a measured or observed quantity, and is never interpreted here as observed human comfort. For each outdoor asset the exposure value is the mean UTCI within a 10 m buffer of the asset's representative point, a spatial statistic fixed in advance; a buffer mean rather than a single centre pixel is used because single-pixel radiant values are sensitive to shadow-edge alignment. Modelled UTCI values are mapped to the same three feasibility states used by the baseline through a pre-registered rule (UTCI ≥ 46 °C → NOT RECOMMENDED; 32 ≤ UTCI < 46 °C → FEASIBLE WITH CONDITIONS; UTCI < 32 °C → FEASIBLE), collapsing the official "strong" and "very strong" stress bands into a single conditional state for direct comparability with the baseline's three-state structure. Indoor assets are not assigned a UTCI value; SOLWEIG models the outdoor environment only, and indoor assets retain their indoor-refuge state.

The modelled Tmrt and UTCI fields are not field-validated anywhere in this study; the station observations validate only the air-temperature, humidity, and wind inputs, not the radiant output. The physical model is therefore treated as a radiation-resolving estimate whose ranges are consistent with published SOLWEIG-class values for comparable climates (e.g. Gál & Kántor, 2019), and never as ground truth or as a more accurate representation of comfort than the baseline.

2.5 Thermal-method comparison

The proxy and physical classifications were compared as a paired reclassification analysis over the 42 outdoor asset × timestamp rows. For each row, the baseline feasibility state and the physically derived feasibility state were placed side by side, and the row was flagged as reclassified where the two states differ. Reclassifications were characterised by direction — whether the physically derived state was more restrictive or less restrictive than the proxy state — and were stratified by timestamp to locate where in the day the two methods agree or diverge. Indoor rows are excluded from this comparison by construction, since they carry no modelled UTCI.

This analysis measures the sensitivity of the feasibility decision to the choice of thermal method, not the accuracy of either method against ground truth: neither method is field-validated, so a reclassified row indicates that the two methods disagree, not that one is correct and the other wrong.

2.6 Uncertainty and robustness treatment

Rather than attach a false-precision error bar to each modelled UTCI value, we carried an evidence-derived uncertainty envelope built only from realizations that were actually computed. For every outdoor row the envelope spans the minimum and maximum UTCI across four solar-forcing realizations — the clear-sky baseline, a satellite-derived irradiance realization (EUMETSAT CM SAF), and −10% and −20% irradiance perturbations — plus, for the two assets whose LiDAR canopy was found stale against a newer per-tree inventory, three corrected-canopy geometry variants bracketing the one crown-radius assumption. The envelope is reported as measured rather than forced to a symmetric band: clear-sky irradiance sits near the physical upper bound of surface radiation and canopy correction only adds shade, so both tested sources widen it mainly below the baseline.

Each row's envelope is translated into a categorical decision-confidence class against the safety-critical 46 °C feasibility boundary (and a secondary 32 °C boundary), using the row's own demonstrated sensitivity — the largest absolute deviation of any realization from the baseline — as the reference margin rather than a fixed tolerance. A row is UNSTABLE if any realization falls on the opposite side of the 46 °C boundary from the baseline; BOUNDARY if no realization crosses but the envelope lies within the row's own demonstrated sensitivity of a threshold; and ROBUST otherwise. Thermal-stress state and decision confidence are kept as separate fields and are never collapsed into a single score: the official UTCI stress category is reported alongside, not multiplied by, the confidence class.

This treatment covers tested solar-forcing and targeted geometry uncertainty only; it does not propagate air-temperature, humidity, wind, or model-structural uncertainty, and there is no field validation of Tmrt or UTCI. A ROBUST label therefore means robust against every uncertainty actually tested, not certain, and the per-row sensitivity is a lower bound on true uncertainty.

2.7 Constraint-first tourism opportunity screening

The screening layer evaluates, for a given source asset at a given hour, which other pilot assets remain feasible candidate alternatives. Each candidate passes through a fixed, ordered sequence of hard constraints, and the first failing constraint determines its outcome; a candidate that clears every constraint becomes a candidate alternative. The order is: pilot scope, then open at the timestamp, then within the walking radius, then within the outdoor thermal limit (UTCI < 46 °C), then evidence sufficiency, and finally a source-relative test of whether the candidate is a meaningful thermal improvement over the source. Cheap categorical eliminations are applied first and the heat-adaptive improvement test last, so that a candidate excluded for being closed is never additionally mislabelled as thermally inadequate. There is no weighted composite score anywhere in the chain (Fig. 1b).

Every candidate carries one machine-readable exclusion reason from a fixed vocabulary (for example, closed at timestamp, accessibility constraint, thermal limit exceeded, insufficient evidence, no meaningful thermal improvement, or outdoor exposure too high), so that each exclusion is traceable to a single most-fundamental cause (Table 2). Thermal state, decision confidence, and evidence confidence are retained as independent fields, with evidence confidence set to the weakest link of opening-hours completeness and thermal-evidence quality; a low value triggers an insufficient-evidence exclusion, so an uncertain thermal decision propagates into the screening outcome rather than being hidden. The meaningful-improvement margin is 0.8 °C — the median per-row demonstrated UTCI sensitivity across the outdoor rows — applied together with categorical rules (an indoor refuge relative to an outdoor source, a strictly cooler outdoor UTCI category, or a confidence gain without becoming hotter).

Surviving candidates are returned as a set annotated with distance, walk time, indoor/outdoor status, experience type, UTCI and its envelope, and the confidence fields, with trade-offs exposed rather than resolved into a single ranked number. The output is a set of screened, surviving alternatives; the framework makes no claim that any tourist will choose, follow, or prefer them.

2.8 Conventional baseline and decision scenarios

To test whether heat-aware screening changes the option set a conventional tool would return, we compare against a deliberately naive nearest-open baseline: the closest open tourism asset within the same walking radius, selected with no thermal or evidence screening. Accessibility for both the screening layer and this baseline is a straight-line (haversine) walking distance converted to time at 4.8 km/h, with a primary pedestrian reach of 800 m (about ten minutes) and sensitivity re-runs at 500 and 1200 m. Straight-line distance is used only as an ordinary-reach constraint and is a documented lower bound on true walking distance; no routing or route-level heat exposure is modelled.

The comparison is run over eight pre-registered decision scenarios, each a real source asset at one of the three hours with a specified accessibility radius, chosen to span the required decision classes — an exposed monument screened to an indoor refuge, an exposed monument screened to a cooler outdoor space, cases with several alternatives of differing experience type, a case where a candidate is available but its evidence is too weak, and a case where the source's own thermal decision is uncertain. One scenario (a source in an already-relatively-cool central park under a constrained 500 m radius) is included as a valid no-survivor test, in which the engine is expected to return an explicit no-recommendation outcome rather than a forced pick. This comparison evaluates whether heat-aware screening changes decision support relative to a proximity-only tool; it does not measure, and makes no claim about, tourist behaviour or travel substitution.

2.9 Validation strategy and claim boundaries

Validation is separated into internal checks, physical-plausibility checks, and what was not validated.

Internal and technical validation. All reported outputs regenerate from the open pipeline over locked inputs with per-file integrity checks; every excluded candidate resolves to exactly one machine-readable reason with no untraceable rows; and the solar-forcing and accessibility-radius sensitivity analyses test the stability of the decisions actually reported.

Physical-model plausibility. The forcing is drawn from real station observations, with the single estimated input (irradiance) gated on an observed clear-sky condition, and the modelled Tmrt and UTCI ranges and their spatial and temporal behaviour were checked for consistency with published SOLWEIG-class values for comparable climates. This is a plausibility check, not a validation of the modelled radiant field.

What was not validated. There is no direct field measurement of Tmrt or UTCI anywhere in this study; no tourist behavioural response, travel substitution, or observed outcome is measured; accessibility is straight-line only, with no route- level heat exposure; and indoor assets are treated as refuges without verified air-conditioning or queue-exposure conditions.

Taken together, these boundaries define what the study evaluates: the sensitivity of tourism-feasibility decisions to the choice of thermal method, and the behaviour of a constraint-first, uncertainty-aware screening architecture relative to a conventional baseline. It does not evaluate the predictive accuracy of either thermal method against ground truth, nor any causal effect on tourism outcomes.

---

3. Results

Results are reported for the locked pilot: 27 tourism assets on 21 August 2023 at 12:00, 15:00, and 18:00, with thermal-method comparison on the 42 outdoor asset × timestamp observations (14 outdoor assets × 3 hours) and screening evaluated across eight pre-registered decision scenarios. All figures are descriptive; the unit is the asset × timestamp observation or the scenario comparison, not a population sample, and no inferential test is applied.

3.1 Thermal-method sensitivity

Across the 42 outdoor observations, the simple-proxy baseline and the physically based (SOLWEIG/UTCI) configuration assigned the same feasibility state in 28 observations and differed in 14, a reclassification rate of 33.3% (14/42) (Table S1). Both directions of change occurred: the physically based configuration produced a more restrictive feasibility state than the proxy in 9 observations (the proxy state was less cautious) and a less restrictive state in 5 (Fig. 2). Reclassification was therefore not one-directional.

The divergence was concentrated in time rather than distributed evenly across the day (Fig. 2). Reclassification reached 64.3% (9/14) at 12:00, fell to 0.0% (0/14) at 15:00, and was 35.7% (5/14) at 18:00. The two methods agreed on every outdoor observation at 15:00 and disagreed most often at noon.

The same rate did not vary with asset morphology. Reclassification was 33.3% within each of the four morphology groups — attraction exterior (3/9), park or garden (6/18), plaza or hardscape (4/12), and street corridor (1/3) — matching the overall rate in every group (Table S1). The time-of-day pattern above was therefore not reproduced across morphology classes.

The noon pattern reflects the difference between the two inputs. At 12:00 the observed ambient air temperature was 34.2 °C, which falls in the LOW meteorological hazard state under the locked AEMET operational warning thresholds. Despite that LOW hazard state, the modelled UTCI equalled or exceeded the 32 °C boundary used in the physical-model feasibility configuration at all 14 of the 14 outdoor assets (Fig. S1). The proxy, keyed to the ambient hazard band, and the physically based configuration, keyed to modelled UTCI, therefore assigned different feasibility states at most noon observations, whereas at 15:00 both configurations placed every outdoor observation in the same conditional-feasibility state.

The finding of this subsection is limited to the following: feasibility classification outcomes were sensitive to the choice of thermal method, and the sensitivity was largest at 12:00.

3.2 Effects on tourism-opportunity screening

Each scenario began from a real asset in a constrained state on the study day. Seven of the eight sources were open outdoor sites in the very-strong-heat-stress category (modelled source UTCI ranging from 38.6 to 45.4 °C) and one (S5, Real Observatorio de Madrid) was closed at its timestamp. Screening behaviour was evaluated across the eight scenarios by comparing the constraint-first surviving-alternative set against a conventional nearest-open baseline that applies no thermal or evidence screening (Table 3). The surviving candidate set differed from the nearest-open set in 7 of the 8 scenarios; it was unchanged in one (S7), where the nearest-open pick already satisfied the locked constraints and no additional candidate was screened out.

The nearest-open pick itself did not satisfy the locked screening constraints in 3 of the 8 scenarios (S2, S6, S8). In each of these three the nearest open asset was an outdoor site that was not cooler than the source, and each was screened out with the same machine-readable reason, `OUTDOOR_EXPOSURE_TOO_HIGH` (Table 3). Across all eight scenarios, 23 open, in-radius candidate options were screened out on thermal or evidence grounds — that is, they were open and within the walking radius but did not satisfy the thermal or evidence constraints (Fig. 3). Surviving alternatives are reported as a set with their trade-offs (indoor/outdoor, distance, experience type, UTCI, and confidence) rather than ranked into a single value.

Seven of the eight scenarios returned a non-empty surviving set (state `ALTERNATIVES_FOUND`), with the number of surviving alternatives ranging from 4 to 9; the eighth returned none. The surviving sets spanned different experience types within a single scenario — for example indoor cultural refuges, transit refuges, and cooler outdoor spaces — which were reported as distinct options rather than merged.

One scenario returned no surviving alternative. In S8 the source was Parque del Retiro at 15:00 under a 500 m accessibility condition. Of the 26 candidate rows evaluated, 0 satisfied the full constraint sequence: the accessible outdoor candidates were not cooler than the already-relatively-cool source and were screened out as `OUTDOOR_EXPOSURE_TOO_HIGH`, while the indoor refuges fell outside the 500 m radius and were screened out as ACCESSIBILITY_CONSTRAINT. The scenario therefore resolved to the explicit state NO_DEFENSIBLE_ALTERNATIVE (Fig. 3). This outcome was produced by the locked screening rules operating on the locked inputs; it was not a manually imposed result or a processing error, and no surviving alternative was withheld.

The finding of this subsection is limited to the following: thermal and evidence screening altered the candidate set relative to a nearest-open baseline in seven scenarios, removed the nearest-open pick in three, and returned an explicit no-alternative state in one.

3.3 Robustness and traceability

The reclassification and screening results were examined against the tested solar-forcing perturbations (Fig. 4). Replacing the clear-sky irradiance estimate with the satellite-derived irradiance realization changed 1 of the 42 outdoor decisions (2.4%). That single change was La Rosaleda (A24) at 18:00, where the satellite realization raised the envelope to the 46 °C feasibility boundary; it is the same observation classified UNSTABLE below. The −10% irradiance perturbation changed 0 of 42 decisions, and the −20% irradiance perturbation also changed 0 of 42. The noon result of Section 3.1 persisted under every tested solar-forcing scenario: all 14 outdoor assets remained at or above the 32 °C UTCI boundary at 12:00 under the clear-sky baseline, the satellite-derived realization, and both irradiance reductions.

Decision confidence was carried as a field separate from thermal state, so that the physiological stress category and the confidence in the feasibility decision were reported side by side rather than combined. Over the 42 outdoor observations the confidence classification was ROBUST for 35 (83.3%), BOUNDARY for 6 (14.3%), and UNSTABLE for 1 (2.4%). The single UNSTABLE observation was La Rosaleda (A24) at 18:00, where a tested realization placed the decision on the opposite side of the 46 °C feasibility boundary from the baseline. The 6 BOUNDARY observations occurred at 15:00 and 18:00, with envelope maxima between 44.5 and 45.9 °C — within their demonstrated sensitivity of the 46 °C boundary but not crossing it. All remaining observations either did not approach a threshold within their demonstrated sensitivity or did not cross one under any tested realization.

The UNSTABLE observation propagated into screening through the evidence gate rather than being suppressed. Where A24 was a candidate (S4), its low evidence confidence produced an INSUFFICIENT_EVIDENCE exclusion while other alternatives survived; where A24 was the source (S7), the screening still returned surviving alternatives and flagged the source's own decision as uncertain. Across the scenarios every excluded candidate carried exactly one machine-readable exclusion reason, and the no-recommendation state (S8) remained an available explicit outcome of the same rule set.

The finding of this subsection is limited to the following: the reported decisions remained unchanged under the tested solar-forcing perturbations with the single noted exception, decision confidence was reported separately from thermal state, and every exclusion and the no-recommendation outcome were traceable to an explicit reason.

4. Discussion

The pilot establishes a single, bounded claim: on this case, different thermal representations can materially change which tourism opportunities a screening system treats as feasible, and a constraint-first architecture can make those changes, the constraints behind them, and their confidence explicit. It does not establish that the physically based configuration is correct and the proxy wrong; neither representation was field-validated here, and the interpretation below is kept to method sensitivity rather than accuracy throughout.

4.1 Thermal representation is decision-relevant

Roughly one-third of outdoor observations changed feasibility state when the thermal input was switched from the simple proxy to the physically based configuration, and the change ran in both directions — the physical configuration was more restrictive in nine observations and less restrictive in five. Two properties of this result matter for interpretation. First, because the reclassification is two-directional, it cannot be read as a one-way bias that a single offset would remove; the choice of thermal representation reshapes decisions in both directions, which is what makes it a decision variable rather than a calibration constant. Second, the divergence is strongly time-dependent — largest at midday, absent at 15:00, intermediate at 18:00 — so it is a feature of when the two representations carry different information, not a uniform gap between them.

The noon pattern makes the mechanism concrete. At 12:00 the observed ambient air temperature was 34.2 °C, which sits in the LOW band of the AEMET civil-protection warning scale, while the modelled UTCI was at or above the 32 °C strong-heat-stress boundary at all fourteen outdoor assets. The two methods encode different thermal information: the operational proxy is keyed to ambient air-temperature warning thresholds and local tree presence, whereas the physical configuration additionally resolves radiant exposure through Tmrt and UTCI. Differences between them can therefore emerge before an air-temperature warning threshold is crossed, precisely when radiant load is high but the air is not yet at a warning level. This is consistent with the urban-microclimate literature in which shading and geometry, not air temperature, dominate mean radiant temperature (e.g. Gál & Kántor, 2019; Ding et al., 2024), and with work arguing that surface- or air-temperature summaries are not interchangeable with human-centred heat-stress indices (Wang et al., 2026).

It is important to state what this does not mean. The AEMET warning thresholds and the UTCI configuration are not two competing measurements of the same quantity. The warning scale is a civil-protection instrument defined on ambient air temperature; UTCI is a modelled index of radiant-inclusive heat load on a standardised body. That a LOW warning state coexists with a strong-heat-stress modelled index does not show that the warning scale understated conditions or that the proxy failed to detect real heat; it shows that the two instruments represent different constructs and can diverge. The contribution here is to quantify that this divergence propagates into tourism-screening decisions, which the composite tourism-climate-index tradition — where weight and threshold choices are already known to move suitability outcomes (Scott et al., 2016; Dubois et al., 2016) — had not examined at the level of a screening decision. The sensitivity to representation is not confined to the proxy-versus-physical contrast: within this study, even simple open-data vegetation proxies did not converge closely with one another, which is consistent with treating the representation of thermal exposure as a substantive decision input rather than a settled preprocessing detail. The practical reading is that thermal representation should be treated as a substantive modelling choice in spatial tourism decision-support, not merely a technical preprocessing step; the pilot supports this on one case rather than as a universal conclusion.

4.2 Screening before routing: a distinct decision-support layer

Heat-aware screening changed the option set that a conventional proximity tool would return. The surviving candidate set differed from a nearest-open baseline in seven of eight scenarios, the nearest-open pick itself did not satisfy the screening constraints in three, and across the scenarios a number of open, in-radius options were removed on thermal or evidence grounds. These are statements about the option set, not about what any visitor would do; no behavioural response is claimed or tested.

The result is best positioned by making the decision-support layer explicit. Heat-aware routing takes an origin and a destination as given and asks which path minimises thermal exposure (Buo et al., 2026; Wolf et al., 2025). Thermal-adjusted accessibility takes a destination set as given and asks how many destinations remain reachable under heat (Aydin et al., 2026). The approach here operates upstream of both: before a route or a reachability count, it asks which tourism opportunities should remain eligible at a given hour under thermal, operational, accessibility, and evidence constraints. Among the closest approaches identified in the targeted review, these routing and accessibility methods presuppose the candidate set, whereas the constraint-first approach produces and filters it. This is a complementary upstream layer rather than a replacement: it does not eliminate the overlap with the routing literature, and a downstream routing or accessibility method could in principle consume the surviving candidate set. Routing was deliberately outside the scope of this study, and the boundary is drawn to complement that frontier, not to compete with it.

The three cases in which the nearest-open pick did not survive screening — each an outdoor site that was not cooler than the source — illustrate a concrete failure mode of proximity-only guidance under heat, and connect to the tourism heat- governance literature that frames attraction-level warnings and closures as management levers (Scott, 2026). The screening layer adds an operational step to that governance picture; it does not, on this evidence, show that using it changes any realised management or visitor outcome.

4.3 Explicit non-recommendation and traceability

One scenario returned no surviving alternative. From Parque del Retiro at 15:00 under a 500 m accessibility condition, all twenty-six evaluated candidates were screened out — accessible outdoor options because they were not cooler than the already-relatively-cool source, indoor refuges because they lay outside the radius — and the scenario resolved to an explicit no-alternative state. This is an architectural property rather than a normative recommendation. A system built on hard, ordered constraints does not need to return a ranked pick when nothing satisfies its predefined conditions, which contrasts with recommender designs that necessarily surface an ordered list or a top choice. The interpretive point is narrow: an explicit no-alternative outcome preserves the meaning of the constraints and prevents a weak option from being presented merely because it is the least unsuitable candidate available.

This behaviour sits alongside equity-oriented accessibility work in which spatial proximity alone is shown to be insufficient for meaningful access to cool refuges (Mombelli et al., 2025), and alongside uncertainty-aware spatial decision-support that treats "no defensible option" as a legitimate result rather than a failure. The traceability of the outcome follows from the same structure: because each candidate is eliminated by the first constraint it fails, every exclusion carries a single machine-readable reason, and the no-alternative state is itself a reason-bearing output rather than a silent gap. This is a property of the first-failing-gate design and of keeping the decision out of a single composite score; it is an architectural and transparency distinction, not empirical proof that the resulting guidance improves management outcomes.

4.4 Robustness, uncertainty, and appropriate confidence

The main decision patterns were stable under the solar-forcing perturbations actually tested. Substituting a satellite-derived irradiance realization for the clear-sky estimate changed one of forty-two decisions, and ±10% and ±20% irradiance perturbations changed none; the noon result held under every tested realization. This is a statement about stability under the tested forcing, not a validation of the modelled radiant field, which remains unmeasured.

The design keeps thermal state and decision confidence as separate fields, and the robustness results show why that separation is useful. A high thermal-stress state can coexist with genuine uncertainty about the precise decision boundary. The one observation classified UNSTABLE — La Rosaleda at 18:00 — is an example: it is firmly in a very-strong-heat-stress state, yet a tested realization moves it across the 46 °C feasibility boundary, so the decision is confidently "hot" while the feasibility call at that specific threshold is not robust. Collapsing stress and confidence into one number would hide exactly this distinction; reporting them separately lets the uncertain case be flagged and, through the evidence gate, excluded when it is a candidate rather than silently carried. The remaining near-boundary observations behave similarly, sitting close to the 46 °C threshold within their own demonstrated sensitivity without crossing it.

This treatment is deliberately partial. Uncertainty was propagated only across the tested dimensions — solar forcing across all outdoor rows and targeted canopy geometry for two assets — and does not cover air-temperature, humidity, wind, or model-structural uncertainty. The confidence labels therefore mean robust against what was tested, not certain. Relative to uncertainty-aware spatial decision-support that propagates input uncertainty into a combined reliability estimate (Sargazi et al., 2025; Huck et al., 2025), the approach here differs by keeping uncertainty categorical and attached to the individual decision rather than folded into one composite figure — a transparency choice, not a claim of more complete quantification.

4.5 Contribution and transferability

Taken together, these results point to an applied decision-support contribution rather than a new metric or algorithm. The proposed framework keeps four things visible as distinct inputs to a screening decision — the thermal representation, the operational and accessibility constraints, the sufficiency of the underlying evidence, and the tested uncertainty — instead of collapsing them into a single composite suitability score. The Madrid pilot is a case-based demonstration that this separation is not cosmetic: changing one of those inputs, the thermal representation, materially altered the feasible candidate set, and the architecture made that change and its confidence auditable. The reproducible pipeline behind the reported tables is part of this contribution; a read-only visual prototype exists as a supplementary demonstration of implementation feasibility and is not itself a result.

Transferability should be read at two levels. The architecture — constraint-first elimination with a first-failing-gate reason, explicit exclusions, confidence kept separate from thermal state, an explicit method-sensitivity comparison, and an explicit no-survivor state — is potentially transferable to other destinations and episodes. Its operational parameters are not automatically transferable: the hazard and UTCI thresholds, the accessibility distances, the curated assets and their opening schedules, the local urban geometry and thermal forcing, and the specific Madrid results are all setting-dependent. The architecture is therefore potentially transferable, whereas its operational parameters require local calibration and validation before use elsewhere.

These interpretations rest on real boundaries that constrain how far they reach. There is no direct field validation of Tmrt or UTCI; the evidence is one city, one extreme-heat day, and twenty-seven assets; the canopy geometry is dated for part of the domain; the uncertainty envelope is partial; accessibility is straight-line with no en-route thermal exposure; the indoor-refuge assumption is unverified for air-conditioning and approach exposure; and the opening hours are documented at a later date than the study day. These limitations bound the interpretation to a demonstration of method sensitivity and of a transparent screening architecture on a single well-characterised pilot; they are set out in full in the following section.

---

5. Limitations and scope conditions

The findings should be read against a set of boundaries that fix what the study can and cannot support. These are stated in order of consequence and are not argued away; several are permanent properties of the design rather than gaps a minor extension would close.

5.1 Physical-model validation boundary

The most consequential boundary concerns the status of the physical thermal fields. Mean radiant temperature and UTCI were model-derived at every asset; no direct field measurements of Tmrt or UTCI were collected anywhere in the study. The available station observations (air temperature, humidity, wind, pressure) support only the meteorological forcing of the model, not the modelled radiant field: they constrain the inputs, not the output. SOLWEIG was assessed through plausibility checks and sensitivity analysis, not against in-situ radiant or comfort measurements, and consistency with published SOLWEIG-class value ranges is a plausibility statement, not a validation of this implementation — precedent for the model in general does not certify the specific configuration used here.

It follows that the proxy-versus-physical comparison evaluates decision sensitivity to the choice of thermal method, not the accuracy of either method against ground truth. A reclassified observation shows that the two representations disagree in a way that changes the screening decision; it does not show that one representation is correct and the other wrong. This boundary is load-bearing for the whole paper: the contribution is the demonstration that method choice is decision-relevant, and it does not require, or assert, that the physical configuration is the more accurate account of on-site conditions.

5.2 Spatial and temporal scope

The empirical evidence is a single case. It covers one city, one pilot area of approximately 3.5 km^2, one documented extreme-heat day, three fixed timestamps, 27 curated tourism assets, and 42 outdoor asset × timestamp comparisons. The assets were purposively selected to span a range of adaptation conditions rather than sampled at random, so the set is not a statistically representative sample of Madrid tourism sites, and the reported rates are properties of this configuration rather than population estimates.

Consequently the study cannot establish seasonal behaviour, climatological averages, city-wide performance across Madrid, cross-city generalisability, or behaviour under different meteorological regimes; a single extreme-heat day says nothing about milder days or other synoptic conditions. This design is appropriate for demonstrating an architecture and for quantifying method sensitivity on a well-characterised case; it is not a basis for population-level or climatological inference, and none is drawn.

5.3 Input and uncertainty limitations

The uncertainty treatment is partial by construction. The reported confidence classes rest on an envelope built from the realizations actually computed: solar forcing across all outdoor rows (a clear-sky estimate, a satellite-derived realization, and −10% and −20% irradiance perturbations) and, for two assets, targeted vegetation-geometry variants. Several uncertainty sources were not propagated — spatial variability in wind, humidity and air-temperature uncertainty, full vegetation uncertainty across the domain, model-structural uncertainty, and the range of possible radiation errors; wind and humidity in particular were applied as single station values across the raster rather than resolved spatially.

The vegetation geometry also carries a temporal-vintage limitation: the LiDAR canopy is dated for parts of the study area, and the targeted corrections addressed the identified decision-critical cases rather than reconstructing the entire urban canopy, so some residual canopy uncertainty remains outside the two corrected sites.

The confidence labels must be read accordingly. A ROBUST, BOUNDARY, or UNSTABLE classification expresses robustness to the tested uncertainty space, not total certainty; the per-row demonstrated sensitivity underlying these labels is a lower bound on the true uncertainty, because it reflects only the dimensions that were varied. A ROBUST label means robust against what was tested, not validated.

5.4 Tourism and accessibility limitations

Accessibility was represented as straight-line distance converted to walking time. The study used no pedestrian route geometry, modelled no thermal exposure during travel, and included no slope or walking-cost model; straight-line distance is a lower bound on true walking distance and an ordinary-reach constraint only. The screening layer therefore operates on candidate opportunities and does not represent the journey to them.

The framework is likewise bounded away from behaviour. It used no behavioural response data, observed no visitor substitution, and was not validated against revealed or stated preferences. It screens the set of candidate opportunities under explicit constraints; it does not predict what tourists will choose, whether they will follow a surviving alternative, whether visitor flows would redistribute, or whether exposure or health outcomes would improve. Those quantities are outside the evidence and are not claimed.

Indoor opportunities require particular caution. Indoor status was handled through a refuge-bypass logic that assumes thermal buffering; indoor air temperature, air conditioning, queue and entrance-waiting exposure, and actual indoor comfort were not observed, and indoor thermal evidence was capped below the highest confidence level by design for this reason. Surviving indoor alternatives should therefore be read as assumed refuges under this logic, not as physically verified cool refuges.

5.5 Operational-data and retrospective-alignment limitations

Opening hours were captured and documented at a later date and applied retrospectively to the 2023 study day; where machine-readable tags were absent, hours were filled from documented institutional schedules, each recorded with its source and a completeness flag. This is sufficient for a reproducible screening demonstration but is not proof that every establishment kept the same schedule on 21 August 2023 — the real Monday-in-August closures used are cited, but a specific establishment could have deviated — and the contextual open-data layers carry their own vintage and completeness limitations of the same kind. This boundary affects the interpretation of a specific candidate's availability at a given hour, not the thermal modelling, which is driven by the observed meteorology for the study day.

5.6 Scope conditions and transferability

These boundaries separate what may transfer from what does not. The potentially transferable part is the architecture: constraint-first screening, first-failing-gate explanations, thermal state kept separate from decision confidence, explicit machine-readable exclusion reasons, an explicit no-survivor state, and the direct comparison of alternative thermal representations. Its operational components are setting-specific and do not transfer automatically: the thermal thresholds, the accessibility radius, the opening-hours schedules, the asset set and its tourism-relevance judgements, the local urban geometry, the meteorological forcing, and the local adaptation resources. The architecture is therefore potentially transferable, whereas its operational parameters and empirical performance require local calibration and validation before use in another setting.

Taken together, these limitations bound the interpretation to a demonstration, on a single well-characterised pilot, that thermal-method choice is decision-relevant for heat-adaptive tourism screening and that a constraint-first architecture can expose that sensitivity, its constraints, and its confidence.

5.7 Future work

The clearest extensions are those that would raise the evidence ceiling rather than add features. Direct field measurement of Tmrt and UTCI would move the physical configuration from plausibility toward accuracy assessment and is the single highest-value next step. Multi-day and seasonal evaluation, and additional city contexts, would test whether the method-sensitivity pattern holds beyond one extreme day and one urban core. Route-level thermal exposure would replace the straight-line reach constraint with a modelled journey; observed tourist behaviour would test whether screening relates to any realised choice; and verified indoor thermal conditions would replace the refuge assumption with measurement. Each targets a specific boundary named above; none is required for the present claim, and none is assumed here.

---

6. Conclusion

This study examined whether the choice of thermal representation affects time-specific screening of urban tourism opportunities under extreme heat, and whether a constraint-first architecture changes the set of feasible alternatives a conventional proximity tool would return. On a single documented extreme-heat day in central Madrid, changing the thermal representation from a simple operational proxy to a physically based (SOLWEIG/UTCI) configuration altered one third of outdoor asset-time feasibility classifications, and the change ran in both directions and was concentrated at particular hours rather than spread evenly through the day. Because it moves in both directions, the divergence is a matter of thermal-method sensitivity, not of one representation being validated against the other; neither was field-measured here. Relative to a conventional nearest-open baseline, the constraint-first screening changed the feasible-alternative set in seven of eight decision scenarios.

The architecture is what makes these differences legible. It keeps the thermal state, the implied decision state, the confidence in that decision, the sufficiency of the underlying evidence, and the reason for each exclusion as separate fields rather than folding them into a single composite suitability score. Because each candidate is eliminated by the first constraint it fails, every exclusion carries a single traceable reason, and the architecture preserves an explicit no-defensible-alternative state for cases in which no accessible option satisfies the constraints rather than returning a weak option by default. This screening step sits upstream of route or reachability calculations: it produces and filters the candidate set that downstream heat-aware routing and accessibility methods presuppose, and it complements rather than replaces them.

Within these bounds, the implication for building heat-adaptive tourism decision-support is that thermal representation should be treated as a substantive modelling choice rather than a technical preprocessing detail. That implication is drawn from a single Madrid pilot, without field validation of the modelled Tmrt or UTCI and without any observation of tourist behaviour, and it is bounded accordingly: the reported rates are properties of this case, not general performance. Multi-day and multi-site evaluation, direct thermal measurement, and observed visitor behaviour would be needed to test how far the pattern and the architecture transfer and whether they relate to any realised outcome. For destination management, the practical point is not that one thermal representation is universally preferable, but that the representation chosen can materially shape which tourism opportunities remain feasible under heat, and should therefore be made explicit, testable, and uncertainty-aware.

Declarations

Funding. [FUNDING STATEMENT TO VERIFY — list any grant/support, or state "This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors."]

Declaration of competing interest. [COMPETING-INTEREST STATEMENT TO VERIFY — e.g. "The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper."]

Data availability. This study uses only open data: meteorological observations from AEMET (Madrid/Barajas station); three-dimensional urban geometry from Spanish national LiDAR products (IGN/CNIG); and tourism assets, tree points, park/garden polygons, and opening-hours tags from OpenStreetMap (ODbL). The derived analytical tables and figures reported here are produced by the project pipeline from these locked inputs. [PUBLIC REPOSITORY / DOI FOR DERIVED OUTPUTS TO VERIFY — a persistent public archive URL/DOI has not yet been established; until then, availability is "on reasonable request."]

Code availability. The screening pipeline and figure-rendering scripts are held in the project repository. [PUBLIC CODE REPOSITORY URL / LICENCE / RELEASE DOI TO VERIFY.]

Author contributions (CRediT). [CREDIT ROLES TO VERIFY per author — e.g. Conceptualization; Methodology; Software; Formal analysis; Data curation; Writing – original draft; Writing – review & editing; Visualization.]

Acknowledgements. [ACKNOWLEDGEMENTS TO VERIFY, or “Not applicable.”]

Ethics. This study involved no human participants, no personal data, and no human-subject research; it analyses open environmental and open geographic data only. No ethics approval was therefore required. [CONFIRM against the target journal’s policy.]

Generative-AI disclosure. [PUBLISHER-POLICY PLACEHOLDER — the target publisher requires authors to disclose any use of generative-AI tools in preparing the manuscript. The authors’ actual tool-use disclosure is to be inserted here verbatim once decided; it is not drafted on the authors’ behalf.]

References

Verified author–year list (23 references). All metadata confirmed from authoritative sources; see *docs/PHASE5_4B1_REFERENCE_VERIFICATION.csv*.

1. Aydin, E. E., Chen, Z., Ortner, F. P., Tay, J. Z., Li, K., & Chua, R. (2026). Universal Thermal Climate Index (UTCI)-adjusted pedestrian accessibility: Urban design exploration for climate-resilience in tropical climates. Sustainable Cities and Society, 143, 107335. https://doi.org/10.1016/j.scs.2026.107335

2. Bröde, P., Fiala, D., Błażejczyk, K., Holmér, I., Jendritzky, G., Kampmann, B., Tinz, B., & Havenith, G. (2012). Deriving the operational procedure for the Universal Thermal Climate Index (UTCI). International Journal of Biometeorology, 56(3), 481-494. https://doi.org/10.1007/s00484-011-0454-1

3. Bunruamkaew, K., & Murayama, Y. (2011). Site suitability evaluation for ecotourism using GIS & AHP: A case study of Surat Thani Province, Thailand. Procedia - Social and Behavioral Sciences, 21, 269-278. https://doi.org/10.1016/j.sbspro.2011.07.02

4. Buo, I., Khan, W. H., Crabtree, E., Emmott, F., Hariyani, D., & Middel, A. (2026). Cool routes: Real-time human thermal exposure routing. Building and Environment, 298, 114622. https://doi.org/10.1016/j.buildenv.2026.114622

5. Camatti, N., Hrast Essenfelder, A., & Giove, S. (2024). Mapping the exposure of tourism to weather extremes: the need for a spatially-explicit gridded dataset for disaster risk reduction. Environmental Research Letters, 19(6), 064008. https://doi.org/10.1088/1748-9326/ad3e91

6. Colaninno, N., Basu, R., Hosseini, M., Alhassan, A., Liu, L., & Sevtsuk, A. (2025). A sidewalk-level urban heat risk assessment framework using pedestrian mobility and urban microclimate modeling. Environment

and Planning B: Urban Analytics and City Science, 52(5), 1071-1090. https://doi.org/10.1177/23998083241280746

7. Ding, X., Zhao, Y., Strebel, D., Fan, Y., Ge, J., & Carmeliet, J. (2024). A WRF-UCM-SOLWEIG framework for mapping thermal comfort and quantifying urban climate drivers: Advancing spatial and temporal resolutions at city scale. Sustainable Cities and Society, 112, 105628. https://doi.org/10.1016/j.scs.2024.105628

8. Dubois, G., Ceron, J. P., Dubois, C., Frías, M. D., & Herrera, S. (2016). Reliability and usability of tourism climate indices. Earth Perspectives, 3(1), 2. https://doi.org/10.1186/s40322-016-0034-y

9. Gössling, S., & Scott, D. (2025). Tourist demand and destination development under climate change: complexities and perspectives. Journal of Sustainable Tourism, 34(5), 1193-1223. https://doi.org/10.1080/09669582.2025.2543953

10. Gál, C. V., & Kántor, N. (2019). Modeling mean radiant temperature in outdoor spaces, A comparative numerical simulation and validation study. Urban Climate, 32, 100571. https://doi.org/10.1016/j.uclim.2019.100571

11. Huck, J. J., Denwood, T., & Taylor, J. E. (2025). Decision making under uncertainty: increasing the impact of public Participatory GIS. International Journal of Geographical Information Science, 40(1), 237-257. https://doi.org/10.1080/13658816.2025.2518556

12. Karimi, A., & Mohammad, P. (2022). Effect of outdoor thermal comfort condition on visit of tourists in historical urban plazas of Sevilla and Madrid. Environmental Science and Pollution Research, 29(40), 60641-60661. https://doi.org/10.1007/s11356-022-20058-8

13. Ketter, E., & Farkash, D. (2026). Tourism in a warming climate: tourist experiences and adaptive responses to rising temperatures in Southern Europe destinations. Sustainability, 18(7), 3454. https://doi.org/10.3390/su18073454

14. Kovács, A., Molnár, G., & Megyeri-Korotaj, O. A. (2025). Projected climate suitability for Hungarian tourism in the 21st century: application of the Holiday Climate Index and modified Tourism Climate Index. International Journal of Biometeorology, 69(6), 1429-1442. https://doi.org/10.1007/s00484-025-02901-y

15. Lindberg, F., Holmer, B., & Thorsson, S. (2008). SOLWEIG 1.0 – Modelling spatial variations of 3D radiant fluxes and mean radiant temperature in complex urban settings. International Journal of Biometeorology, 52(7), 697-713. https://doi.org/10.1007/s00484-008-0162-7

16. Mombelli, S., González March, R., Cucchietti, F., Marquet, O., & Reyes, P. (2025). Are Barcelona’s climate shelters accessible to vulnerable residents? A mobility justice analysis. Cities, 168, 106487. https://doi.org/10.1016/j.cities.2025.106487

17. OECD (2026). OECD Tourism Trends and Policies 2026. OECD Publishing, Paris. https://doi.org/10.1787/3fd3cd75-en

18. Renninger, A., & Cabrera, C. (2026). Extreme heat reduces and reshapes urban mobility. PNAS Nexus, 5(4), pgag078. https://doi.org/10.1093/pnasnexus/pgag078

19. Sargazi, M. A., Heidari, A., Davtalab, J., & Piri, J. (2025). Comparative reliability assessment of PET and UTCI thermal comfort indices using Monte Carlo simulation in urban microclimates. Scientific Reports, 16, 3431. https://doi.org/10.1038/s41598-025-33440-6

20. Scott, D. (2026). A framework for heat risk action planning for tourism. Annals of Tourism Research, 118, 104141. https://doi.org/10.1016/j.annals.2026.104141

21. Scott, D., Rutty, M., Amelung, B., & Tang, M. (2016). An inter-comparison of the Holiday Climate Index (HCI) and the Tourism Climate Index (TCI) in Europe. Atmosphere, 7(6), 80. https://doi.org/10.3390/atmos7060080

22. Wang, Y., Yi, S., Li, X., Liu, P., Yang, Z., Bardhan, R., & Stouffs, R. (2026). From physical surfaces to human-centric heat stress: LST and UTCI heat mapping reveals nonlinear effects of urban morphology. arXiv preprint arXiv:2604.22433. https://arxiv.org/abs/2604.22433

23. Wolf, H., Vierø, A. R., & Szell, M. (2025). CoolWalks for active mobility in urban street networks. Scientific Reports, 15, 14911. https://doi.org/10.1038/s41598-025-97200-2

Figure captions

Figure 1. Study design and constraint-first screening architecture. (a) The Prado–Retiro–Atocha Madrid pilot study area (approximately 3.5 km² in central Madrid), with the 27 curated tourism assets marked as outdoor (n = 14) or indoor (n = 13); the extent is a bounded pilot and does not represent city-wide Madrid. (b) The analytical architecture: 27 tourism assets are screened with two alternative thermal representations — a simple operational proxy (ambient air-temperature hazard band + nearby tree presence) and a physically based model (SOLWEIG → Tmrt → UTCI) — through one ordered, constraint-first gate chain (open? → reachable? → thermally feasible? → evidence sufficient? → meaningful improvement?), returning either surviving alternatives or an explicit no-defensible-alternative outcome. Thermal state, decision confidence, evidence confidence, and exclusion reason are kept as separate fields; no composite score is used.

Figure 2. Thermal-method choice changes feasibility classifications, in both directions. Unit of analysis: the outdoor asset × timestamp observation (14 outdoor assets × 3 timestamps = 42 observations). (a) Agreement matrix; each cell is one of three states — agreement (both methods assign the same feasibility state), physical configuration more restrictive than the proxy, or physical configuration less restrictive than the proxy. “More restrictive” and “less restrictive” describe the direction of classification divergence between the two methods, not the correctness of either; neither method is field-validated. (b) Direction of divergence by timestamp as a share of the 14 outdoor assets, with the reclassification rate per timestamp (12:00 = 64.3%, 15:00 = 0.0%, 18:00 = 35.7%). Overall, 14/42 observations (33.3%) were reclassified — 9 physical more restrictive, 5 physical less restrictive, 28 in agreement.

Figure 3. Heat-aware screening changes the option set relative to a conventional nearest-open baseline. Eight pre-registered decision scenarios (S1–S8),

each a real source asset at one timestamp and walking reach. For each scenario: the nearest-open baseline pick and whether it passes the constraint-first screen (● passes; ○ removed, with the machine-readable exclusion reason); the number of surviving alternatives (0–9); and the count of open, in-radius options removed on thermal or evidence grounds. Across the eight scenarios the candidate set changed in 7 of 8, the nearest-open pick was removed by screening in 3 of 8 (each `OUTDOOR_EXPOSURE_TOO_HIGH`), and 23 open, in-radius options were removed in total. S8 (outlined; Parque del Retiro, 15:00, 500 m reach) evaluated 26 candidates, of which 0 survived, resolving to `NO_DEFENSIBLE_ALTERNATIVE`; this is an architectural outcome of the locked constraints, not a ranked recommendation.

Figure 4. Decision robustness under tested uncertainty. (a) Number of the 42 outdoor decisions that changed under each tested solar-forcing realization relative to the clear-sky baseline: a satellite-derived irradiance realization changed 1/42 (2.4%), and −10% and −20% irradiance perturbations changed 0/42 each. (b) Distribution of the categorical decision-confidence class over the 42 outdoor observations — ROBUST 35 (83.3%), BOUNDARY 6 (14.3%), UNSTABLE 1 (2.4%) — with the single UNSTABLE case (A24, 18:00) at the 46 °C feasibility boundary annotated. Robustness here refers only to the tested uncertainty dimensions (solar forcing across all outdoor rows plus targeted canopy geometry for two assets); it is not total uncertainty and is not a validation of the modelled field.

Table captions

Table 1. Data sources and provenance. Open-data layers used in the analytical chain, their provider, spatial/temporal support, licence, and what each layer does and does not measure. See `manuscript/tables/TABLE01_data_sources.md`.

Table 2. Constraint-first screening architecture. The ordered gate chain, the threshold for each gate, the machine-readable exclusion vocabulary, and the separate decision fields (thermal state, decision confidence, evidence confidence, exclusion reason). See `manuscript/tables/TABLE02_screening_rules.md`.

Table 3. Scenario comparison: constraint-first screening versus nearest-open baseline (S1–S8). For each scenario, the source asset, timestamp and reach, the nearest-open baseline pick and whether it survives screening, the number of surviving alternatives, the count removed on thermal/evidence grounds, and the outcome state. See `manuscript/tables/TABLE03_scenarios.md`.

Supplementary Figure S1 and Supplementary Tables S1–S4 are described in *supplementary/SUPPLEMENTARY_MATERIAL_v0.1.md*.